STEVENS INSTITUTE OF TECHNOLOGY

Data Acquisition and Processing I: Big Data

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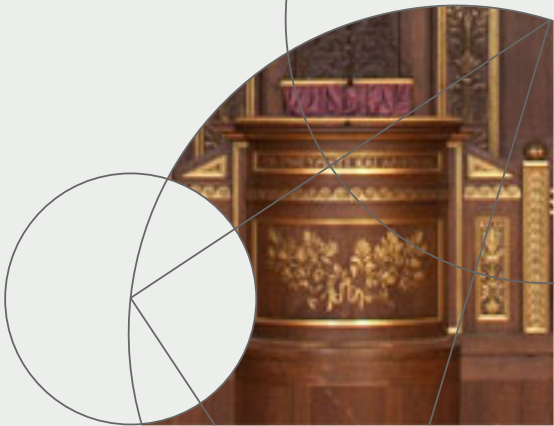

ARIMA Model

- A popular and widely used statistical method for time series forecasting is the ARIMA model.
- ARIMA is an acronym that stands for AutoRegressive Integrated Moving Average. It is a class of model that captures a suite of different standard temporal structures in time series data.
- This acronym is descriptive, capturing the key aspects of the model itself.
  - **AR**: Autoregression. A model that uses the dependent relationship between an observation and some number of lagged observations.
  - **I**: Integrated. The use of differencing of raw observations (e.g. subtracting an observation from an observation at the previous time step) in order to make the time series stationary.
  - **MA**: Moving Average. A model that uses the dependency between an observation and a residual error from a moving average model applied to lagged observations.

ARIMA Model

- Each of these components are explicitly specified in the model as a parameter. A standard notation is used of ARIMA(p, d, q) where the parameters are substituted with integer values to quickly indicate the specific ARIMA model being used.
- The parameters of the ARIMA model are defined as follows:
  - p: The number of lag observations included in the model, also called the lag order.
  - d: The number of times that the raw observations are differenced, also called the degree of differencing.
  - q: The size of the moving average window, also called the order of moving average.

ARIMA Model: What is integrated d?

- d is the number of times that the raw observations are differenced.
- For example, we have the raw observations $\{x_t\}$.
- For $d = 1$, it means, you have a first difference

$$y_t = x_t - x_{t-1}$$

Then y_t is fitted for ARMA model (not ARIMA), i.e.,

$$y_t = a_0 + \sum_{i=1}^{p} a_i y_{t-i} + \epsilon_t + \sum_{j=1}^{q} b_j \epsilon_{t-j}$$

- For $d = 2$, you need to have two differences:

$$y_t = x_t - x_{t-1} \quad \text{and} \quad z_t = y_t - y_{t-1}$$

Then z_t is fitted for ARMA model

$$z_t = a_0 + \sum_{i=1}^{p} a_i z_{t-i} + \epsilon_t + \sum_{j=1}^{q} b_j \epsilon_{t-j}$$

ARIMA Model: What is zero for p, d, q?

- If the $p = 0$ in the set of (p, d, q), the ARIMA(p,d,q) reduces to a model with integrated and moving average model (NO auto-regressive).
- If the $d = 0$ in the set of (p, d, q), the ARIMA(p,d,q) reduces to a standard ARMA model with auto-regressive order p and moving average order q.
- If the $q = 0$ in the set of (p, d, q), the ARIMA(p,d,q) reduces to a model with AR(p) after d times differences.
- If the $d = 0$ and $q = 0$ in the set of (p, d, q), the ARIMA(p,d,q) reduces to a standard AR(p) model.
- If the $d = 0$ and $p = 0$ in the set of (p, d, q), the ARIMA(p,d,q) reduces to a standard MA(q) model.

ARIMA Model

- A linear regression model is constructed including the specified number and type of terms, and the data is prepared by a degree of differencing in order to make it stationary, i.e. to remove trend and seasonal structures that negatively affect the regression model.
- A value of 0 can be used for a parameter, which indicates to not use that element of the model. This way, the ARIMA model can be configured to perform the function of an ARMA model, and even a simple AR, I, or MA model.
- Adopting an ARIMA model for a time series assumes that the underlying process that generated the observations is an ARIMA process. This may seem obvious, but helps to motivate the need to confirm the assumptions of the model in the raw observations and in the residual errors of forecasts from the model.

ARIMA Model

- Steps to be followed for ARIMA modeling:
  1. Exploratory analysis
  2. Fit the model
  3. Diagnostic measures
- You need to use all 3 of them iteratively

ARIMA Model

Exploratory analysis

- Make sure the time series is stationary
- Check if there is a trend
- Check if there is a cyclic pattern

ARIMA Model – Model Selection

- The ultimate goal is to if the model has fit all the data except the noise space
- In other words, the best model should have the residual model fitting errors close to the white noise mostly
- Check the residual errors from the fitted model with ACF and PACF
- Finally, compare with different models with the AIC(Akaike information criterion) value, the smallest one is the best option
- AIC value is

$$\text{AIC} = 2k - 2\ln(\hat{L})$$

where k be the number of estimated parameters in the model and $\hat{L}$ be the maximum value of the likelihood function for the model.

ARIMA Model – Model Selection

- AIC estimates the relative amount of information lost by a given model: the less information a model loses, the higher the quality of that model.
- In estimating the amount of information lost by a model, AIC deals with the trade-off between the goodness of fit of the model and the simplicity of the model. In other words, AIC deals with both the risk of overfitting and the risk of underfitting.
- In R, after calling the ARIMA model, use `summary()` to list the estimated parameters in the model and the related AIC value. Again, smaller AIC, better model.

```
testModel <- ARIMA(your_Data_vector, c(p,d,q))
summary(testModel)
testModel$residuals
```

ARIMA Model – What is the `$residuals` in mdoel?

- For example, you have a simple AR(1) model:

$$x_t = a_0 + a_1 x_{t-1} + \epsilon_t$$

  where ϵ_t is the noise term.

- After you run the model ARIMA(1,0,0), you will have a set of estimated model parameters, i.e., $\{\hat{a}_0,\ \hat{a}_1\}$.

- The `$residuals` is the difference between the raw observation and the fitted model

$$\texttt{\$residuals} = x_t - (\hat{a}_0 + \hat{a}_1 x_{t-1})$$

- Although ϵ_t is the noise term which is not observable, you can use the `$residuals` to testify whether your estimated model is good enough.

ARIMA Model – How to testify the `$residuals` in mdoel?

- The criterion is that `$residuals` should be close to a Gaussian noise term, i.e., `$residuals` $\approx \epsilon_t$.
- To testify if `$residuals` is close to the noise term ϵ_t, one can plot the ACF and PACF of `$residuals`.
- An ideal noise term ϵ_t should have both ACF(l)=0 and PACF(l)=0 for all $l > 0$.
- If the above ACF and PACF equal to zero conditions are not observed, one can adjust the model parameters and obtain a new `$residuals` time-series and run ACF and PACF again. This process can be iteratively implemented.